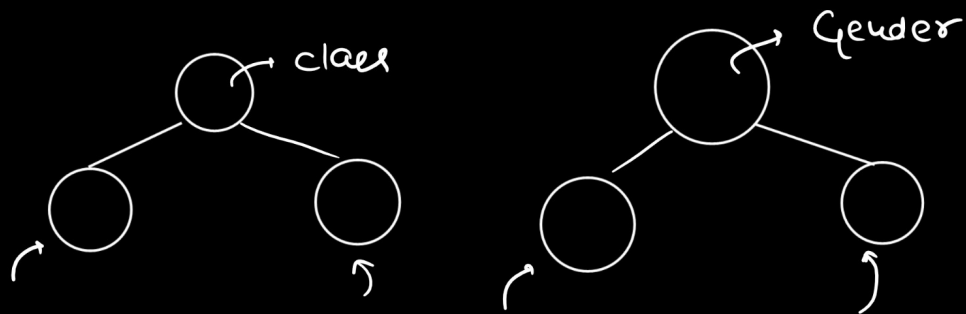


Decision Tree $\rightarrow$ Regression

$$\Downarrow \frac{\text{Sum of squared Error}}{\text{Error}} \Rightarrow \sum_{i \in L} (y_i - \hat{y}_L)^2 + \sum_{i \in R} (y_i - \hat{y}_R)^2$$



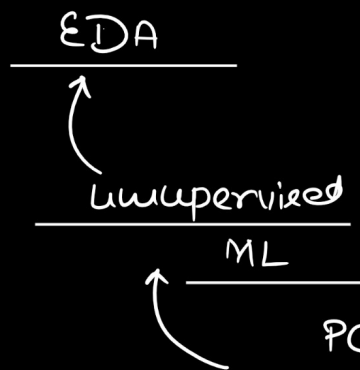
Dimensionality Reduction

Dimensional $\rightarrow$ feature

500 features
50,000 records } $\rightarrow$ data shape

① training becomes very slow

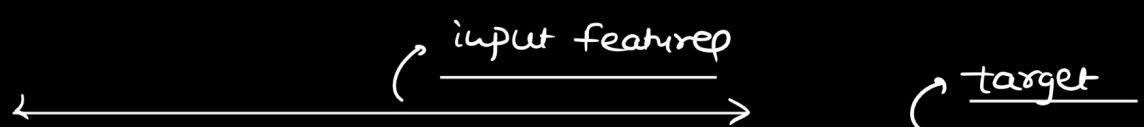
② Visualization of the too many features becomes impossible



PCA $\rightarrow$ Principal Component Analysis

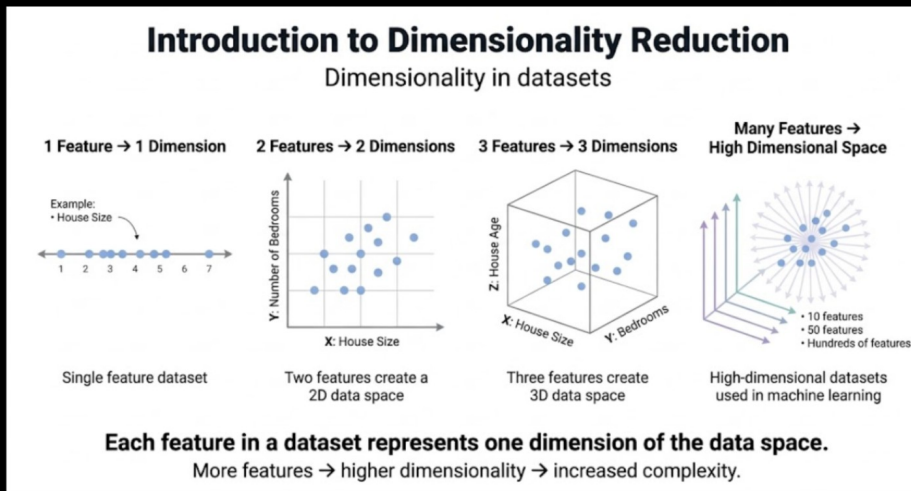
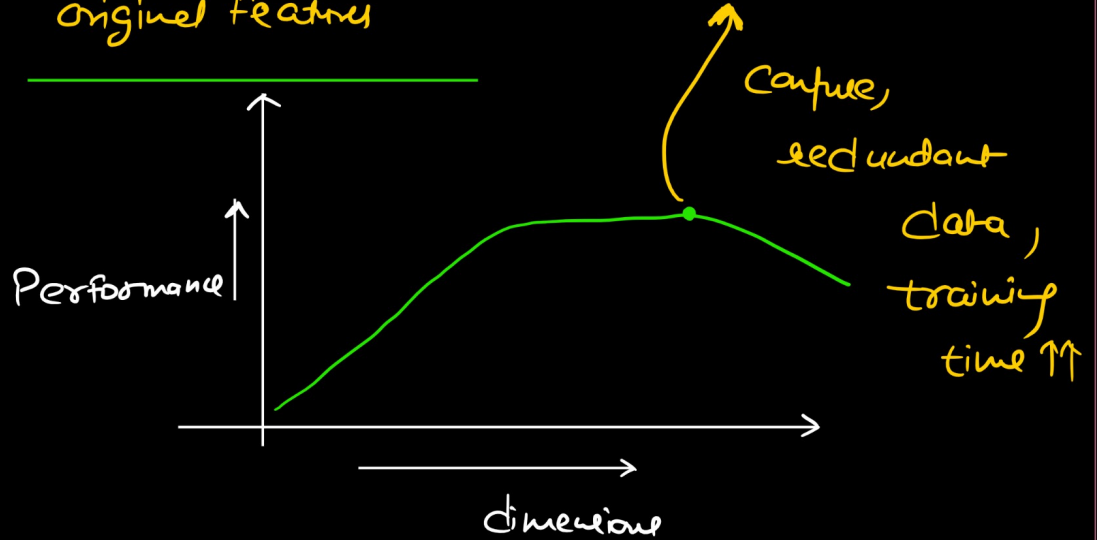
transform original features into a new set of orthogonal features

called as principal components.



$f_1 \quad f_2 \quad f_3 \quad f_4 \quad f_5 \quad \dots \quad f_{500} \quad y$

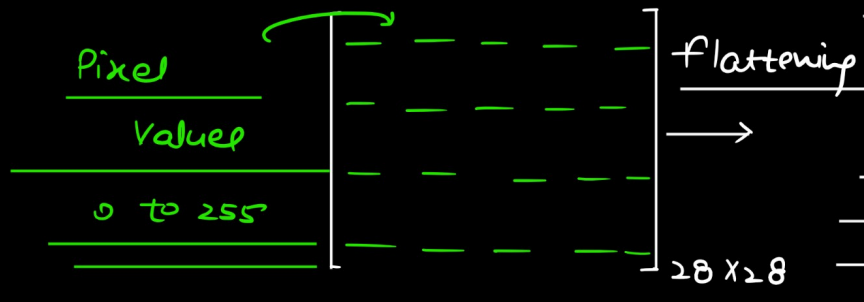
50 new features which
 contains max variation/information
 of the original features
 PCA
 curse of dimensionality



Real time example of high dimensional

Data 28×28
 $\geq 18^4$

Image data → 28×28



↳ Textual data → numeric vector

English

"Machine Learning models analyze the data"

thousands of dimensions

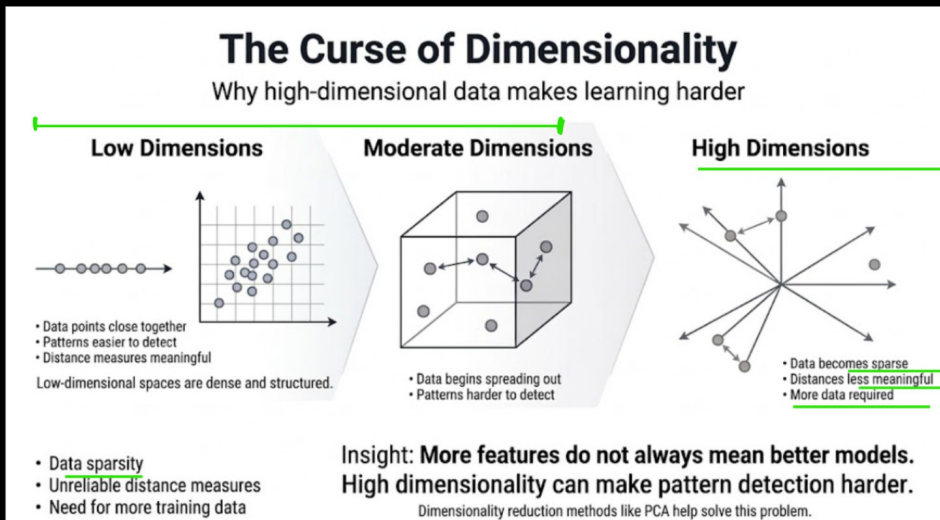
Hindi (Language translation)

↳ Customer Behavior data

↳ age, location, purchases, click activity,

time on website, device type &

many more



↳ having more zeros
in a sparse
matrix

Curse of Dimensionality: As the number of features increases, the feature space becomes increasingly empty, causing distances between data points to lose their meaning and algorithms (like K-nearest neighbors) to degrade

↳ Distance
metric

feature selection

f_1 ~~f_2~~ f_3 ~~f_4~~ f_5 ~~f_6~~ ~~f_7~~ ~~f_8~~

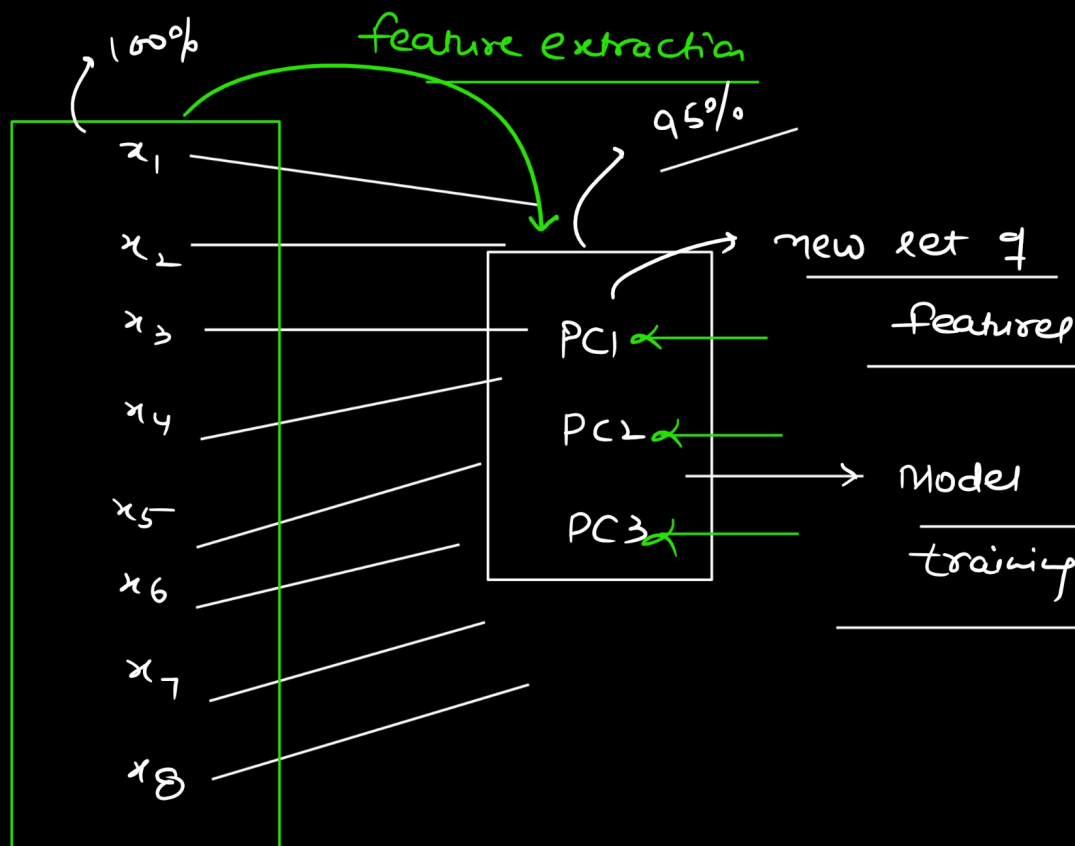
f_1, f_3, f_5

Dimensionality

Reduction

② feature extraction

→ create new features
from existing ones



feature engineering → feature

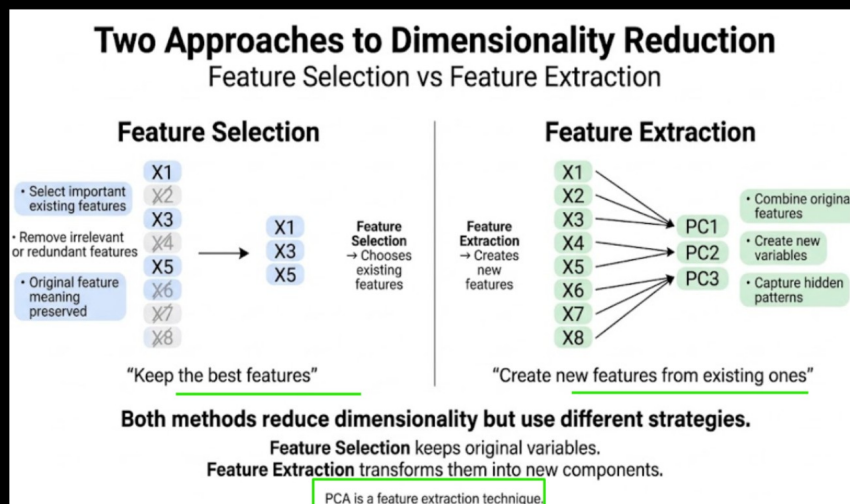
~~date of birth~~

age

selection,
feature
transformation

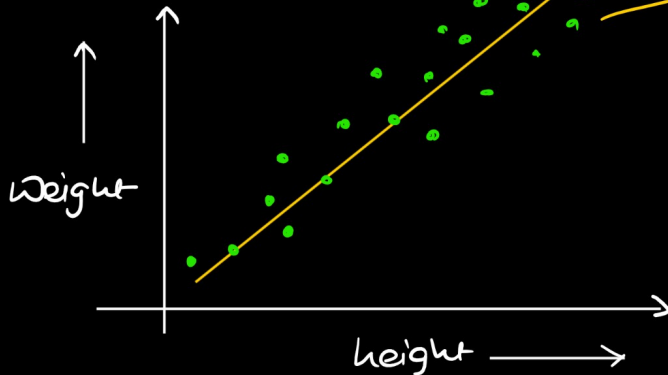
~~firstName, lastName~~

fullName



eigen value
& eigen
vector

2D data



1D data

transformation

rotate axis

eigen vector

(indicates the
property of

both height &

weight)

Steps required to build Principal
Components using eigenvalues &
eigenvectors

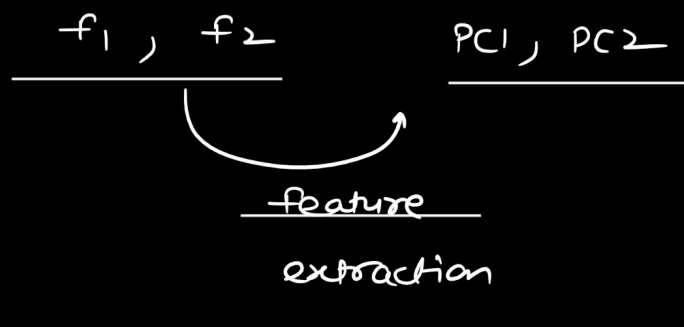
① Small sample data (scratch)

② Real time data & usage
of inbuilt

method of

Principle Components

$$\frac{\text{Covariance matrix}}{\text{matrix}} \Rightarrow \frac{df.T * df}{n-1}$$



maximum info of the
PC1 > PC2 original
feature

$x \rightarrow \text{variance } y$

$x, y \rightarrow \text{input feature}$

$$\text{Cov}(A) = \begin{matrix} \begin{matrix} x \\ y \end{matrix} & \begin{bmatrix} \frac{\sum (x_i - \bar{x})(x_i - \bar{x})}{N} & \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N} \\ \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N} & \frac{\sum (y_i - \bar{y})(y_i - \bar{y})}{N} \end{bmatrix} \end{matrix}$$

(x, y)

variance

$\text{variance} = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$

$\text{variance} \Rightarrow \frac{\sum_{i=1}^n (x_i - \mu)(x_i - \mu)}{n}$

(x)